

Early life compartmentalization of human T cell differentiation and regulatory function in mucosal and lymphoid tissues

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Evidence abstract

It is unclear how the immune response in early life becomes appropriately stimulated to provide protection while also avoiding excessive activation as a result of diverse new antigens.

T cells are integral to adaptive immunity; mouse studies indicate that tissue localization of T cell subsets is important for both protective immunity and immunoregulation.

In humans, however, the early development and function of T cells in tissues remain unexplored.

We present here an analysis of lymphoid and mucosal tissue T cells derived from pediatric organ donors in the first two years of life, as compared to adult organ donors, revealing early compartmentalization of T cell differentiation and regulation.

Whereas adult tissues contain a predominance of memory T cells, in pediatric blood and tissues the main subset consists of naive recent thymic emigrants, with effector memory T cells (T(EM)) found only in the lungs and small intestine.

Additionally, regulatory T (T(reg)) cells comprise a high proportion (30-40%) of CD4(+) T cells in pediatric tissues but are present at much lower frequencies (1-10%) in adult tissues.

Pediatric tissue T(reg) cells suppress endogenous T cell activation, and early T cell functionality is confined to the mucosal sites that have the lowest T(reg):T(EM) cell ratios, which suggests control in situ of immune responses in early life.

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Evidence checklist

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Source continuity

The canonical evidence above remains the authoritative retrieval target.